Solution:

Formula:

$$\bar{X} = \frac{\sum X}{N}$$

$$\bar{X} = \frac{25.3+26.58+28.46+30.61+31.32+30.75+30+29.34+29.03+27.89+26.45+25.36}{12}$$

$$\bar{X} = \frac{341.09}{12}$$

$$\bar{X} = 28.42$$

Answer:

The Mean annual temperature of Cuddalore district is 28.42

Calculation of Mean for Grouped Data or Continuous Series

Formula:

$$\bar{X} = A + \frac{\sum fd}{N} \times i$$

Where

A = Assumed mean
f = frequency
i = class size
d = deviation from assumed mean
N = Total

$$d = \frac{x - A}{i}$$

Example 13.2

Calculate the mean for the following data of heights of the plants in a garden

Heights in cms	10-20	20-30	30-40	40-50	50-60
No. of plants	5	15	10	15	5

Solution:

Heights in cms X	No of Plants F	Mid point x	A = 25, i = 10, $d = \frac{x - A}{i}$	fd
10 – 20	5	15	-1	-5
20 – 30	15	25 A	0	0
30 – 40	10	35	1	10
40 – 50	15	45	2	30
50 – 60	5	55	3	15
	N =50			**Σfd =55-5 = 50**

A = 25 , ∑ fd = 50 , N = 50, i = 10

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